

ANUBHAV AGRAWAL

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TECHNICAL SKILLS

Languages & Tools: Python, Git, Linux, $\text{\LaTeX}$

ML & Deep Learning: PyTorch, OpenCV, Scikit-Learn, XGBoost, NumPy, Pandas

LLMs & RAG: OpenAI API, LangChain, LangGraph, HuggingFace, LLM Guardrails

Inference & Serving: TensorRT, ONNX, CUDA, INT8/FP16 Quantization, FastAPI, vLLM, llama.cpp

Platform & MLOps: AWS (EC2), Docker, GitLab CI/CD, LGTM stack (Grafana, Mimir, Loki, Tempo), MongoDB

EXPERIENCE

Griphic

Delhi

AI Engineer II

Jan 2026 – Present

- Architected and developed, as sole engineer, a production multi-agent LLM service (Python/FastAPI) generating structured interview blueprints end-to-end, processing **10K+ requests per month**.
- Built bounded self-correction loops around schema-validated structured outputs, delivering **99.5%** of generations with zero human review; the LLM repairs its own failures within a fixed retry budget.
- Designed a two-stage guardrail system (batched judge $\rightarrow$ independent adjudicator) against prompt injection, jailbreaks, and leaked credentials; eliminated false-positive rejections of legitimate inputs while holding guard overhead to ~ 2 LLM calls per clean request.
- Cut per-request generation cost by **60%** via token-level cost telemetry, prompt caching, and bounded parallel fan-out.
- Enforced quality gates as code: GitLab CI/CD with unit + BDD suites and SonarQube ($\geq 90\%$ coverage, VAPT/ISO 27001-aligned checks, Trivy HIGH/CRITICAL blocks).

Innvolution Healthcare Pvt. Ltd.

Bangalore

Machine Learning Engineer & Team Lead

Sep 2023 – Dec 2025

- Owned AIMAG, an X-ray super-resolution and denoising product reaching **\$200K+ ARR**, from initial concept and dataset curation through model R&D, quantization, and release after clinical trials.
- Served the model in real time: PyTorch $\rightarrow$ ONNX $\rightarrow$ TensorRT with INT8/FP16 **post-training quantization** — **sub-5 ms per-frame** latency on an RTX 4090 for $4\times$ upsampling.
- Engineered the QCA pipeline (segmentation $\rightarrow$ calibration $\rightarrow$ prediction), **fine-tuning** U-Net++ (ResNet50 + attention), delivering sub-4-second processing for live cardiac-catheterization analysis.
- Co-invented **3 patent applications** in X-ray image enhancement: blind single-image super-resolution, DSA roadmap generation, and real-time fluoroscopy enhancement.
- Led a team of **6 ML engineers** — owned the technical roadmap, instituted architecture and code review practices, and guided solutions from research prototype to production deployment.

EDUCATION

Bhilai Institute of Technology, Durg — B.Tech in Computer Science Engineering

2019 – 2023

PROJECTS

Inference Lab — LLM Serving Benchmarks on Apple Silicon

[GitHub](#) | [Write-up](#)

Reproducible benchmark suite: mlx-lm vs llama.cpp under a single fairness contract

2026

- Built a reproducible benchmark suite holding mlx-lm and llama.cpp to one fairness contract (fixed prompts, greedy decoding, matched budgets): **45–56 tok/s** single-stream for Qwen2.5-7B-Instruct (4-bit) on an M4 Pro.
- Isolated why throughput stays **flat from 1 to 32 concurrent users**: a per-instance lock serializes requests inside each serving engine.
- Proved decode is memory-bandwidth-bound: K process-isolated instances behind one dispatcher return only **+9%** for **3 $\times$ the memory**, since a single instance already saturates the M4 Pro's ~ 273 GB/s bus.

ipic — On-Device Multimodal RAG File Manager (Rust)

[GitHub](#) | [Write-ups](#)

Fully offline search across text, PDFs, images, audio, and video — no cloud, no telemetry

2026

- Architected a sharded index — one self-contained SQLite + vector-store shard per root — under a supervising engine that parallelizes scan, embedding, and search; fully offline multimodal search across text, PDFs, image content, and spoken queries.
- Fused dense (bge-small), CLIP vision, BM25 (FTS5), and acoustic lanes via reciprocal rank fusion over custom mmap'd i8-quantized vector stores — exact scan chosen over HNSW for exact recall at file-index scale: **26–32 ms** queries, sub-millisecond vector scans.
- Scaled to a **350k-file** home directory at **~ 2.5 k files/s** with bounded-channel backpressure, batched shard claims, and a per-stage compute budget — eliminating the SQLite convoy and queue growth that capped v1.